

Abstract

This study investigates whether the association between self-defeating and self-enhancing humor differs significantly between younger and older university students. Data was obtained from the Open Psychometrics database, specifically the Humor Styles Questionnaire (HSQ) developed by Martin et al. (2003), consisting of 989 respondents after cleaning. Respondents were divided into two age groups using a median split: under 23 and 23 or above, resulting in approximately 495 respondents per group. The four humor style scores were recalculated from raw item responses due to inconsistencies found in the dataset's pre-computed scores. Spearman correlation and Fisher's Z test were conducted with self-defeating humor as the independent variable and self-enhancing humor as the dependent variable. Results indicate that both age groups show a statistically significant positive association between the two humor styles, contradicting the original hypothesis, and that this association is significantly stronger in younger respondents than older ones.

Introduction

Humor is not a single, uniform behavior — it comes in different styles that reflect different psychological tendencies. Research by Martin et al. (2003) identified four distinct humor styles: affiliative, self-enhancing, aggressive, and self-defeating. Of these, self-defeating humor — the tendency to put oneself down for laughs or to allow others to ridicule oneself — has been theorized as a psychologically maladaptive style, often associated with low self-esteem and poor emotional coping. Self-enhancing humor, by contrast, is considered adaptive — the ability to find amusement in life's difficulties and maintain a positive outlook through humor even when alone.

The prevailing theoretical expectation is that these two styles operate in opposition: individuals high in self-defeating humor should be low in self-enhancing humor, as the former reflects self-deprecation while the latter reflects healthy self-regulation. This study sets out to test whether that expected negative relationship actually exists in real data, and whether the pattern differs

between younger and older respondents. Age is a meaningful grouping variable here because emotional maturity and self-perception tend to shift across early adulthood, potentially changing how humor styles relate to one another.

The central research question is: *Does the association between self-defeating and self-enhancing humor differ significantly between younger (under 23) and older (23 or above) university students?*

Data

The data used in this study was obtained from the Open Psychometrics public database at openpsychometrics.org, which hosts raw data from interactive online psychological tests. The specific dataset used was collected using an online version of the Humor Styles Questionnaire (HSQ), originally developed by Martin, Puhlik-Doris, Larsen, Gray, and Weir (2003). Respondents completed the 32-item questionnaire voluntarily and consented to having their responses used for research purposes.

The original dataset contained 1,071 respondents across 39 variables, including 32 raw item responses, four pre-computed scale scores, age, gender, and a self-reported accuracy rating. Several data quality issues were identified and addressed during cleaning. First, 78 respondents had at least one missing item response (coded as -1), and these were removed entirely to ensure all scale scores were computed consistently. Second, four respondents reported implausible ages (including values of 44,849, 2,670, 242, and 151), which were removed using a cutoff of age greater than 90. After cleaning, the working sample consisted of 989 respondents.

A median split was applied to the age variable (median age = 23) to create two groups: under 23 (n = 495) and 23 or above (n = 494). Several limitations of this dataset must be acknowledged, including self-reporting bias introduced through the survey format, the voluntary and self-selected nature of the sample, and the fact that the sample cannot be considered globally representative.

Methodology

The four humor style scores — affiliative, self-enhancing, aggressive, and self-defeating — were recalculated from the raw item responses using the original scoring formulas provided in the dataset codebook. This was necessary because manual verification revealed that the pre-computed scores in the dataset could not be exactly reproduced from the raw items using standard scoring methods when missing values were present, suggesting an undocumented handling of missing data by the original data collector. All recalculated scores are averages of eight items each, scored on a 1-5 scale.

For this analysis, self-defeating humor served as the independent variable and self-enhancing humor as the dependent variable. Age group (under 23 vs 23 or above) served as the grouping variable. Two formal statistical tests were conducted. The first was Spearman correlation, chosen because the data is ordinal and does not assume a normal distribution. Spearman correlation measures the strength and direction of the association between the two humor variables within each age group separately. The second test was Fisher's Z test, which compares two independent correlation coefficients to determine whether the difference between them is statistically significant. An alpha level of 0.05 was used for all tests. All analysis was conducted in Python using the pandas, scipy, and matplotlib libraries.

Results

Descriptive statistics revealed that self-defeating humor had a mean score of 2.75 across the full sample, while self-enhancing humor had a mean of 3.38. Both distributions were approximately normal with moderate spread, consistent with moderate humor style tendencies across the sample.

Histograms of self-defeating humor by age group showed right skewed distributions for both groups, with peaks in the 2.6-3.4 bin. The older group showed a slightly larger proportion of respondents at the lower end of self-defeating humor. Histograms of self-enhancing humor showed approximately normal distributions for both groups, with peaks in the 2.6-4.2 range. The older group appeared slightly more concentrated in the middle range.

The scatter plot of self-defeating versus self-enhancing humor showed upward sloping regression lines for both age groups, suggesting a positive rather than negative association. The younger group's regression line was steeper, indicating a stronger association.

Spearman correlation analysis revealed that for respondents under 23, $r = 0.2641$ with $p = 0.0000$ — a weak but highly statistically significant positive correlation. For respondents aged 23 or above, $r = 0.1362$ with $p = 0.0024$ — also statistically significant but weaker. Fisher's Z test confirmed that the difference between these two correlations is statistically significant ($Z = 2.0922$, $p = 0.0364$).

Discussion

The central finding of this study is that self-defeating and self-enhancing humor show a positive association in both age groups — directly contradicting the original hypothesis that they operate in opposition. Rather than replacing healthy coping humor, self-defeating humor appears to coexist with it. People who score high on one style tend to score high on the other.

There are several possible explanations for this unexpected finding. First, humor use in general may be a trait-like tendency — people who engage more with humor of any kind simply use more of all humor styles, regardless of their adaptive or maladaptive nature. Someone who frequently makes self-deprecating jokes may also be someone who frequently uses humor to cope with stress, simply because they are a more humor-oriented person overall. Second, the self-selected nature of this dataset — people voluntarily taking an online humor test — may mean the sample skews toward individuals who are more reflective about and engaged with humor in general, compressing the expected negative relationship between styles.

The finding that younger respondents show a stronger positive association than older respondents is also notable. This suggests that as people age, the two humor styles become more differentiated — older respondents may develop clearer distinctions between adaptive and maladaptive humor use, weakening the tendency for the two styles to move together. This is consistent with broader research on emotional maturity and self-regulation across early adulthood.

It is worth noting that no causation is implied in this analysis. The data is cross-sectional and correlational — it can identify associations but cannot determine what drives them.

Limitations

Several limitations of this study must be acknowledged. The sample is self-selected, consisting of individuals who voluntarily chose to complete an online humor questionnaire, which may not be representative of the broader population. Both humor style scores are self-reported, introducing potential response bias — people may not accurately assess their own humor tendencies. The age grouping used a median split, which is a standard methodological approach but introduces some arbitrariness in the cutoff. The sample sizes, while balanced, were not drawn from a controlled study design. Finally, the dataset's pre-computed scores could not be exactly verified against the raw items, raising minor concerns about the original data processing, though the recalculated scores used in this analysis are fully transparent and reproducible.

Closing Thoughts

This study found a statistically significant positive association between self-defeating and self-enhancing humor in both younger and older respondents, with the association being significantly stronger in the younger group. These findings challenge the theoretical expectation that the two styles operate in opposition, suggesting instead that humor use may be a general trait that cuts across style distinctions, particularly in younger individuals. Future research using longitudinal designs, objective humor assessments, and more diverse samples would help clarify the nature and direction of these relationships.

References

- Martin, R. A., Puhlik-Doris, P., Larsen, G., Gray, J., & Weir, K. (2003). Individual differences in uses of humor and their relation to psychological well-being: Development of the Humor Styles Questionnaire. *Journal of Research in Personality*, 37, 48-75.
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